

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 2

Geometry, Trigonometry & Circles

Duration: 2.5 hours

Total Marks: 100

Name: _____

Date: _____

SECTION A: Short Answer Questions (40 marks)

1. In triangle ABC, angle A = 50° , angle B = 60° . Find angle C. [2 marks]

2. Using sine rule, if $a/\sin A = b/\sin B$, and $a = 10$, $A = 30^\circ$, $B = 45^\circ$, find b. [3 marks]

3. Find the area of triangle with sides 5 cm, 6 cm, 7 cm using Heron's formula. [4 marks]

4. In a circle with radius 5 cm, find arc length if angle at center is 60° . [3 marks]

5. If $\sin \theta = 3/5$ and θ is acute, find $\cos \theta$ and $\tan \theta$. [3 marks]

6. Prove: $\sin^2 \theta + \cos^2 \theta = 1$ [2 marks]

7. Find the bearing from A to B if the angle from north is 120° . [2 marks]

8. In right triangle, hypotenuse = 10, one side = 6. Find other side. [2 marks]

SECTION B: Long Answer Questions (60 marks)

9. Triangle ABC: AB = 8 cm, BC = 6 cm, angle ABC = 60° . Find AC using cosine rule. (5 marks)

10. Two buildings of heights 20m and 30m are 50m apart. Find angle of elevation between their tops. (5 marks)

11. Prove that $\cos(A + B) = \cos A \cos B - \sin A \sin B$ (5 marks)

12. A circle has center O and radius 6 cm. Chord AB is 8 cm. Find distance from O to chord AB. (4 marks)

13. Solve $\sin 2x = 1/2$ for $0 \leq x \leq 180^\circ$. (6 marks)

14. In a quadrilateral ABCD, $AB = 5$, $BC = 6$, $CD = 7$, $DA = 8$, angle $ABC = 90^\circ$. Find diagonal AC. (5 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Find angle C

$$A + B + C = 180^\circ, C = 180^\circ - 50^\circ - 60^\circ = 70^\circ$$

Q2: Sine rule

$$10/\sin 30^\circ = b/\sin 45^\circ, b = 10 \sin 45^\circ/\sin 30^\circ = 10 \times (\sqrt{2}/2) / (1/2) = 10\sqrt{2}$$

Q3: Heron's formula

$$s = (5+6+7)/2 = 9, \text{Area} = \sqrt{(9 \times 4 \times 3 \times 2)} = \sqrt{216} = 6\sqrt{6} \text{ cm}^2$$

Q4: Arc length

$$\text{Arc} = r\theta \text{ (}\theta \text{ in radians)} = 5 \times \pi/3 = 5\pi/3 \text{ cm}$$

Q5: Find cos and tan

$$\sin^2\theta + \cos^2\theta = 1, (3/5)^2 + \cos^2\theta = 1, \cos^2\theta = 16/25, \cos \theta = 4/5, \tan \theta = 3/4$$

Q6: Prove identity

$$\text{Using Pythagoras on unit circle with point } (\cos \theta, \sin \theta) \text{ on circle: } \cos^2\theta + \sin^2\theta = 1$$

Q7: Bearing

Bearing is 120° (measured clockwise from north)

Q8: Pythagoras

$$a^2 + 6^2 = 10^2, a^2 = 100 - 36 = 64, a = 8$$

SECTION B SOLUTIONS

Q9: Cosine rule

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(ABC), AC^2 = 64 + 36 - 2(8)(6)\cos 60^\circ, AC^2 = 100 - 48, \\ AC = \sqrt{52} = 2\sqrt{13} \text{ cm}$$

Q10: Angle elevation

$$\text{Height difference} = 10\text{m}, \text{horizontal distance} = 50\text{m}, \tan(\text{angle}) = 10/50 = 1/5, \text{angle} = \arctan(1/5) \approx 11.3^\circ$$

Q11: Prove cos(A+B) identity

Use angle addition formula or geometric proof with unit circle

Q12: Distance to chord

Distance d: $d^2 + 4^2 = 6^2$, $d^2 = 36 - 16 = 20$, $d = 2\sqrt{5}$ cm

Q13: Solve $\sin 2x = 1/2$

$2x = 30^\circ$ or 150° or 390° or 510° , $x = 15^\circ, 75^\circ, 195^\circ, 255^\circ$

Q14: Find AC

Using Pythagoras in right triangle ABC: $AC^2 = AB^2 + BC^2 = 25 + 36 = 61$, $AC = \sqrt{61}$